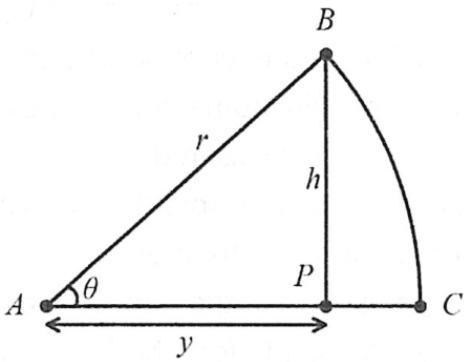
Trigonometry: Small Angle Approximations

Questions 1–24

Question 1. The diagram below shows a sector of a circle, ABC. By considering trigonometric ratios, show that as $\theta \to 0$ that $\sin\theta \approx \tan\theta \approx \theta$



Question 2. Show that if $\sin\theta \approx \theta$ then as $\theta \to 0$, $\cos\theta \approx 1 - \frac{\theta^2}{2}$

Question 3. When θ is small, find an approximation for the following:

a) $\theta \sin \theta$
b) $\cos \theta - 1$
c) $\cos 6\theta$
d) $\tan \theta - 2 \sin \theta$
e) $\theta \tan 3\theta + \cos 3\theta$

Question 4. When θ is small enough for θ^3 to be ignored, find an approximation for the following:

a) $\frac{\theta \tan \theta}{1 - \cos \theta}$
b) $\cos \theta \cos 3\theta$
c) $\cos^2 \theta$
d) $\sin^2 \theta + \cos^2 \theta$
e) $\frac{\theta^2 \sin \theta}{1 - \cos 2\theta}$
f) $\frac{\cos 8\theta - \cos 4\theta}{\sin 8\theta - \sin 4\theta}$
g) $\frac{1 - \cos 2\theta}{6 \tan \theta}$

Question 5.

a) Find an approximation for $\cos 0.4$ by using $\theta = 0.1$ and small angle approximations.

b) Find the percentage error of your approximation.

Question 6. When θ is small, use the standard small angle approximations for $\sin\theta$ and $\tan\theta$ find:

a) $\theta \operatorname{cosec}\theta$

b) $\cot 2\theta$

c) $\cot 4\theta - \operatorname{cosec} 2\theta$

Question 7.

$$f(x) = \sec x(\sin x + \cos x)$$

a) By using the small angle approximations for $\tan x$ show that $f(x) \approx 1 + x$

b) By using the small angle approximation for $\sin x$ and an alternative approximation, $\cos x \approx 1$, show that $f(x) \approx 1 + x$

c) Find the percentage error when approximating $f(0.1)$

Question 8.

a) When θ is small, show that $-2 + \cos\theta + \sin^2\theta \approx -1 + \frac{1}{2}\theta^2$

b) Hence find an approximate solution to $-2 + \cos\theta + \sin^2\theta = 4\tan\theta$

Question 9. Given x is small, find the approximate solution to

$$\frac{1-\cos(\frac{1}{10}x)}{\sin x} = \frac{1}{5} - \tan x$$

Question 10.

a) Find, for small values of x, an approximation for $\mathrm{f}(x) = 2 + 4\tan\frac{x}{2} - 8\sin\frac{x}{3}$

b) Use your result in part a) to sketch an approximation for the graph of $\mathrm{f}(x)$ for $-0.2 \leq x \leq 0.2$. Include the start and end coordinates in your sketch.

Question 11. Given that the root of the equation is small, find the single solution to $2x - \frac{1}{4} + \cos x = 0$

Question 12. Given x is small enough for x^3 to be ignored, by approximating $\tan x + \tan x \cos 4x$ show that

$$\int_0^{0.3} \sqrt[3]{\tan x(1 + \cos 4x)}\, dx \approx 0.190$$

Question 13.

a) When θ is small, find an approximation to $\cos\theta - \cos 2\theta$

b) Hence find $\lim_{\theta\to 0} \frac{\cos\theta - \cos 2\theta}{\theta^2}$

Question 14. Find these limits:

a) $\lim_{h\to 0} \frac{\cos h - 1}{h}$

b) $\lim_{h\to 0} \frac{\sin h}{h}$

Question 15. Prerequisite knowledge for Q15-Q18: Differentiation from First Principles and Compound Angle Formulae Differentiate from first principles the following functions by using knowledge about small angle approximations:

a) $\mathrm{f}(\theta) = \sin\theta$

b) $\mathrm{f}(\theta) = \cos\theta$

Question 16. Using differentiation from first principles, for $\mathrm{f}(x) = \cos(x)$, find the exact value of $\mathrm{f}'(\frac{\pi}{3})$

Question 17. Prove, using differentiation from first principles, that $\frac{d}{dx}(\tan\theta) = \sec^2\theta$

Question 18. Prove, using differentiation from first principles, that $\frac{d}{dx}(\sec\theta) = \sec\theta\tan\theta$

Question 19. Prerequisite knowledge for Q19-Q20: Binomial Series

a) Find the first three terms, in ascending powers of x of the binomial expansion of $(1-8x)^{1/2}$

b) Hence, for small values of x, show that $\sin 2x + \sqrt{\cos 4x} \approx a + bx + cx^2$ where a, b and c are constants to be found and x is small enough for x^4 to be ignored.

Question 20.

a) Find the first two terms, in ascending powers of x of the binomial expansion of $(1-\frac{1}{2}x^2)^{1/3}$

b) Hence find an approximation for $\int_0^{0.2} \sqrt[3]{\cos x}\, dx$ giving your answer to 4 decimal places.

Question 21. Prerequisite knowledge for Q21-Q22: Differentiation of Trigonometric Functions Find the value of $\frac{dy}{dx}$ at the origin of $x = \frac{1}{2}\tan 4y$ by:

a) using Calculus

b) using a small angle approximation and Calculus

Question 22. Find the value of $\frac{dy}{dx}$ when $x = 0$ for $x = 3\cos 16y$ where $0 < y < \frac{\pi}{2}$ by:

a) using Calculus

b) using a small angle approximation and Calculus

Question 23. Prerequisite knowledge for Q23-Q24: Newton-Raphson Method Given x is small, find an approximate solution giving your answers to 3 significant figures for $\frac{1}{2} - x - \sin 2x = 0$

a) by using small angle approximation

b) by using a Newton-Raphson approximation starting at $x_0 = 0$

Question 24. Given x is small, find an approximate solution giving your answers to 3 significant figures for $2 + 5x + \sin x = \cos x$

a) by using small angle approximation

b) by using a Newton-Raphson approximation starting at $x_0 = 0$